

MA 572: Applied Multivariate Statistical Analysis

Course Code	MA 572
Course Name	Applied Multivariate Statistical Analysis
L-T-P-C	3-0-2-4
Prerequisites	Probability and Statistics, Linear Algebra, Computer Programming
Intended for	B.Tech 4th year, M.Tech/MSc, Ph.D.
Distribution	Elective
Approval	59th BoA (Apr 2025)

Course Contents

- **Recap of Prerequisites:** A quick overview of matrix computations, random variables, probability distributions & estimations. [4 Lectures]
- **Introduction to Multivariate Analysis:** Definition and scope of multivariate analysis: from univariate to multivariate thinking and the importance of data exploration; Applications in real life scenarios; Organization of multivariate data; Data cleaning and preprocessing (handling missing values, outliers, scaling, transformations); Covariance and correlation matrices (geometric interpretation); Mahalanobis distance (geometric interpretation and outlier detection); Data visualization techniques for multivariate data (scatter plots, pair plots, heatmaps, parallel coordinate plots, Andrews curves, Chernoff faces); Real-world examples. [6 Lectures]
- **Multivariate Distributions:** Multivariate normal distribution: Definition, properties (marginal and conditional distributions, linear combinations), probability density function, geometric interpretation, and applications in hypothesis testing; Wishart distribution: Definition, properties, and its role in covariance matrix estimation; Hotelling's T^2 distribution: Derivation, properties, and relation to univariate t-distribution; Multivariate t-distribution: Definition, properties, and applications in robust statistical modeling; Other multivariate distributions and their applications. [9 Lectures]
- **Multivariate Regression Models:** Introduction to multivariate regression: Multiple response variables and their modeling: Fitting and estimating multivariate regression models; Model diagnostics: Residual analysis and influence measures; Assessing model fit and significance; Generalized multivariate regression: Extending multivariate regression to handle non-linear relationships; Applications of multivariate regression in practice. [6 Lectures]
- **Canonical Correlation Analysis (CCA):** Mathematical formulation of CCA: Finding linear combinations of variables from two sets that have maximum correlation; Derivation of canonical variates and canonical correlations; interpretation of canonical correlations and canonical weights/loadings; Assumptions and limitations of CCA; Redundancy analysis; Relationship to other multivariate techniques; Applications of CCA in practice. [4 Lectures]
- **Multicollinearity and Regularization Techniques:** Detecting multicollinearity: Variance inflation factor (VIF), correlation matrix; Consequences of multicollinearity on regression models; Regularization methods: Ridge regression and Lasso regression; Tuning parameters and their impact on multicollinearity; Solving multicollinearity issues with regularization. [4 Lectures]
- **Multivariate Analysis of Variance (MANOVA) and Related Techniques:** Analysis of variance (ANOVA): One-way and two-way ANOVA; Assumptions, hypothesis testing; Multivariate analysis of variance (MANOVA): Comparing multiple groups with multiple response variables, assumptions, hypothesis testing, interpretation of results; Different tests such as likelihood ratio tests, etc.; Testing linear hypotheses in multivariate regression models. [6 Lectures]
- **Case Studies and Model Evaluation:** Case studies from different application areas; Model validation techniques: Train-test split, cross-validation, and bootstrapping; Performance evaluation metrics: R-squared, adjusted R-squared, MSE, RMSE, precision, recall, F1-score, ROC curve, and AUC; Residual analysis, outlier detection, and influence diagnostics: Model stability, robustness, and interpretation of coefficients; Strategies for addressing overfitting and underfitting.

Laboratory Modules

The lab component of this course focuses on the practical application of multivariate analysis techniques through hands-on implementation with real datasets. It aims to enhance students' conceptual understanding and develop their ability to apply techniques and interpret results effectively.

Text Books

1. Johnson, R. A. and Wichern, D. W., **Applied multivariate Statistical Analysis**, PHI learning pvt ltd, 2002.
2. Anderson, T. W., **An introduction to Multivariate Statistical Analysis**, Wiley India, 2010.

References

1. Everitt, B. S. and Dunn G., **Applied Multivariate Data Analysis**, Wiley- Blackwell, 2001.
2. Srivastava, M.S., **Methods of Multivariate Statistics**, Wiley- Interscience, 2002.
3. Tan, P. N., Steinbach M., Kumar V., **Introduction to Data Mining**, Pearson, 2014.
4. Flury, B., **A First Course in Multivariate Statistics** (Springer Texts in Statistics), Springer-Verlag, 1997.